President's Office

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

ARUSHA DISTRICT COUNCIL

FORM FOUR PRE-MOCK EXAMINATION - MARCH 2020

041 BASIC MATHEMATICS

Time:3 Hours **Friday, 27th March 2020 a.m**

INSTRUCTIONS:

- This paper consists of section A, B
- Answer **all** questions from section A and B
- NECTA Mathematical tables may be used
- All necessary work and answers for each question must be shown clearly
- Where necessary use $\pi = 3.14$ and radius of the Earth = 6370km
- Cellular phones and any unauthorized materials are not allowed in the examination room.
- Write your **Examination number** on every page of your answer booklet(s)

SECTION A (60 Marks)

Answer **all** questions in this section

1. (a) Find the product of the LCM and GCF of 16, 24, and 36

   (b) Evaluate $3\frac{9}{10} \div \left(3\frac{3}{5} - 1\frac{1}{2}\right)$ and express your answer correct to 2 decimal places

2. (a) Show that $\log\left(\sqrt[3]{x}\right) = \bar{3}.3508$ if $\log x = \bar{8}.0524$

   (b) Rationalize the denominator of $\frac{\sqrt{5}+\sqrt{2}}{\sqrt{6}-\sqrt{2}}$

3. (a) Given that ∪= {1,2,3,4,5,6,7,8,9,10}; A ={2,5,6,8,10} and B = {3,6,7,9};
   find (i) A∩B' (ii) A'∪B'

   (b) John and David are about to sit for their final exams, the probability of John passing the exams is 50%, while the probability of David not-passing the exams is $\frac{1}{4}$; find the probability that

   (i) David will pass the exams

   (ii) Either John will pass the exams or David will fail the exams

4. (a) Find the equation of the line which passes through the point A(-4,3) and which is perpendicular to the line 3x = 15 – 4y

   (b) Given the vectors $\underline{x} = 3i + 2j, \underline{y} = 5i - 3j$ and $\underline{z} = 4i - 2j$; find

   (i) the resultant vector $\underline{r} = \underline{x} + \underline{y} + \underline{z}$ and its direction

   (ii) Magnitude of $\underline{r}$.

5. (a) A regular nonagon is inscribed in a circle of radius 9cm. Find the perimeter and area of this polygon.

   (b) Given that triangle ABC is similar to triangle XYZ where AB = 4cm, BC = 5cm, XY = 18cm and angle XYZ = 30^0; find the area of the triangle XYZ.

6. (a) Given that x is inversely proportional to y and directly proportional to the square of z when y = 4, x = 2 and z = 3; find the value of x when y = 3 and z = 2

   (b)Mr. Xuma from South Africa visited Tanzania with 10000/- Rands and wanted to change his cash to USD. If on that day 1USD was equivalent to 2650/- TSh, and 1Rand was equivalent to 42/- TSh, how much USD did he get?

7. (a) Two numbers are respectively 20% and 50% more than a third number. What is the ratio of the two numbers?

   (b) The following information relates to Kalulu Traders as 31st December, 2019.
   Net Profit 1000000/-
   Cost of Sales 70% of Sales
   Purchases 8000000/-
   Closing stock 30% of Purchases
   Sales to Net profit ratio 20:1
   Determine:
   (i) Sales
   (ii) Cost of Sales

(iii) Closing Stock
(iv) Opening Stock
(v) Expenses

8. (a) The 3rd, 6th, and 12th terms of an AP form the first three terms of a geometric progression. If the difference between the 10th term and the 7th term of A.P is 9. Find the sum of the first five terms of G.P

(b) Paula invested a certain amount of money in a savings bank whose interest rate was 10% compounded annually. After 2 years she got 5000000/= TShs.

(i) How much did she invest at the start?

(ii) How much did she receive as interest at the end of two years

9. (a) Without using mathematical tables or calculators find the value of $\frac{\tan 45^0 - \tan 30^0 \sin 60^0}{\cos 45^0}$

(b) A man who is 2m tall stands on a horizontal ground 30m from a tree. The angle of elevation of the top of the tree from his eye is 28^0. What is the height of the tree?

10. (a) Find the values of t and s in the following system of equations $\begin{cases} 3t + s = 17 \\ 27 - 3t - 6s = 0 \end{cases}$

(b) Given that $a = (3x + \frac{1}{2})$ and $b = (\frac{x}{2} - 3)$, find the value of x if it is known that

$ab + 1 = 1$

SECTION B(40 Marks)

Answer **all** questions in this section

11. (a) The table bellow shows the masses in kg of form one students at Kalakala Secondary School

Class Interval	41 – 45	46 - 50	51 - 55	56 - 60	61 - 65	66 - 70
Frequency	3	8	13	11	7	3

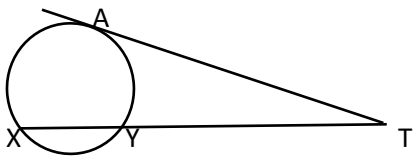
(i) Find the total number of students

(ii) Draw a frequency polygon and histogram on the same axes

(iii) Calculate the mode

(iv) Calculate the Median

(b) Calculate the value of TA in figure bellow leaving your answer in radical form given that XY = 3cm, and YT = 9cm, while A is the tangent point to the circle



12. (a) Find the distance in km between two places on the same meridian with latitudes

(i) 15^0S and 13^0N

(ii) 20^0N and 42^0N

(b) An Airplane flies from town A(5^0S, 33^0E) to town B(5^0S, 39^0E) at 332 km/h along a parallel of latitude . If it leaves town A at 5.00Am, find the arrival time at town B airport.

(c) ABCDV is a square pyramid with the base ABCD while V is vertically above the center E of the base

(i) Draw the three dimensional figure of the pyramid

(ii) Is it true that BV and AD are skew lines? Give reason(s)

13. (a) (i) Given the matrices $A = \begin{pmatrix} -2 & 3 \\ 4 & 5 \end{pmatrix}$ and $B = \begin{pmatrix} -15 & 9 \\ 3 & 6 \end{pmatrix}$, find 4A - $\frac{B}{3}$

(ii) If the matrix $N = \begin{pmatrix} 9n & -2 \\ -8 & 4n \end{pmatrix}$ is singular, find the possible value of n

(b) Solve the following system of linear equations by using the inverse matrix method

$$\begin{cases} 2x = 6 - y \\ -y - 6 = -6x \end{cases}$$

(c) Find the image of the point (-2,5) under a reflection in the line y+x = 0

14. (a) Mr. Samson wants to plant maize and beans. Maize needs 3 men per hectare and beans needs 3 men per hectare. He has 48 hired men. To maintain a hectare of maize he needs TShs. 250000/- while a hectare of beans costs TShs. 100000/-. Find the greatest area of land he can plant if he prepared to use Tsh. 25000000/-

(b) Given that $f(x) = 3x^2-15x+18$, find

(i) Maximum or minimum value of f

(ii) Turning point (iii) f(-4) – f(2)